Measure qubits

Package versions

To get information about a qubit's state, you can *measure* it onto a classical bit. In Qiskit, measurements are performed in the computational basis, that is, the single-qubit Pauli-Z basis. Therefore, a measurement yields 0 or 1, depending on the overlap with the Pauli-Z eigenstates $|0\rangle$ and $|1\rangle$:

$$|q\rangle \xrightarrow{measure} \begin{cases} 0 (\text{outcome} + 1), \text{with probability } p_0 = |\langle q|0\rangle|^2, \\ 1 (\text{outcome} - 1), \text{with probability } p_1 = |\langle q|1\rangle|^2. \end{cases}$$

Apply a measurement to a circuit

There are several ways to apply measurements to a circuit:

`QuantumCircuit.measure` method

Use the `measure` method to measure a `QuantumCircuit`.

Examples:

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(5, 5)
qc.x(0)
qc.x(1)
qc.x(4)
qc.measure(
    range(5), range(5)
)  #  Measures all qubits into the corresponding clbit.
```

Output:

```
<qiskit.circuit.instructionset.InstructionSet at 0x7f2cb8
```

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(3, 1)
qc.x([0, 2])
qc.measure(1, 0)  # Measure qubit 1 into the classical bi
```

Output:

```
<qiskit.circuit.instructionset.InstructionSet at 0x7f2cb8
```

`Measure` class

The Qiskit Measure class measures the specified qubits.

```
from qiskit.circuit import Measure

qc = QuantumCircuit(3, 1)
qc.x([0, 1])
qc.append(Measure(), [0], [0])  # measure qubit 0 into cl
```

Output:

```
<qiskit.circuit.instructionset.InstructionSet at 0x7f2cb8
```

`QuantumCircuit.measure_all` method

To measure all qubits into the corresponding classical bits, use the `measure_all` method. By default, this method adds new classical bits in a `ClassicalRegister` to store these measurements.

```python
from qiskit import QuantumCircuit

qc = QuantumCircuit(3, 1)
qc.x([0, 2])
qc.measure_all()  # Measure all qubits.
```

`QuantumCircuit.measure_active` method

To measure all qubits that are not idle, use the `measure_active` method. This method creates a new `ClassicalRegister` with a size equal to the number of non-idle qubits being measured.

```python
from qiskit import QuantumCircuit

qc = QuantumCircuit(3, 1)
qc.x([0, 2])
qc.measure_active()  # Measure qubits that are not idle,
```

Next steps

Recommendations

- `Measure` class
- `measure_all` method
- `measure_active` method
- `random_circuit` method
- Mid-circuit measurements (Available only when using IBM Quantum Compute Service.)

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